

Algebra II with Trigonometry

Chapter 7.3

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4592201_alg-2trig-h-chp-7-3-note-docx.pdf](#)

Based on the content in the PDF (focusing on double-angle, half-angle, and product-sum formulas), here are **10 mental math questions** that can be answered without paper and pencil. Each question tests understanding of the core trigonometric concepts from the document:

Double-Angle and Half-Angle Questions

1. If $\sin x = 1/2$, what is $\sin 2x$?

- *(Hint: Use the double-angle formula for sine.)*

2. If $\cos x = 0$, what is $\cos 2x$?

- *(Use the double-angle formula for cosine.)*

3. If $\tan x = 1$, what is $\tan 2x$?

- *(Apply the double-angle formula for tangent.)*

4. If $\cos x = 1$, what is $\cos^2(x/2)$?

- *(Use the half-angle formula for cosine squared.)*

Lowering Powers

5. If $\cos 2x = 0$, what is $\sin^2 x$?

- *(Recall the power-reducing formula for sine squared.)*

Product-Sum and Sum-Product Formulas

6. What is $\sin 0 \cdot \cos 0$ expressed as a sum?

- *(Product-to-sum formula for $\sin u \cos v$.)*

7. If $\sin x + \sin x = k$, what is k in terms of $\sin x$?

- *(Sum-to-product simplification.)*

8. Evaluate $\cos x \cdot \cos x$ using a product-to-sum formula. What is the sum you get?

- *(Product-to-sum formula for $\cos u \cos v$, where $u = v = x$.)*

Conceptual & Quick Application

9. What is always true about $\sin 2x$ if $x = 0^\circ$?

- *(Think about the value of sine at zero.)*

10. If $\sin x = 0$, what is $\tan 2x$?

- *(Use the double-angle formula for tangent, remembering $\tan x = 0$.)*

Optional challenge question (for deeper understanding):

- **If $\cos x = 0$, what is $\sin^2 x + \cos^2 x$?**
- *(Test the fundamental identity.)*

Let me know if you'd like quick explanations or answers for each!